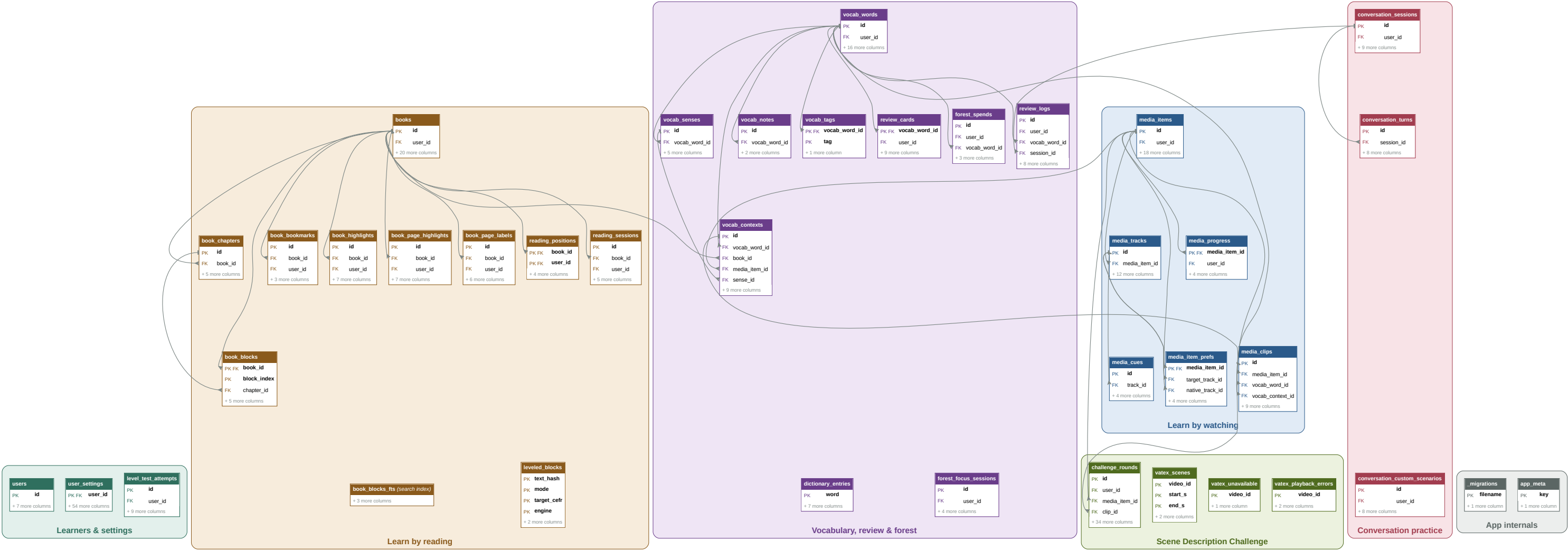


# FluencyOS — database schema

SQLite database `fluencyos.db` · 39 tables · 51 relationships · generated from a freshly migrated database (44 migrations applied). This page shows only key columns; the following pages show every column. Lines to users are left out here to keep it readable — every table with a `user_id` belongs to one learner.

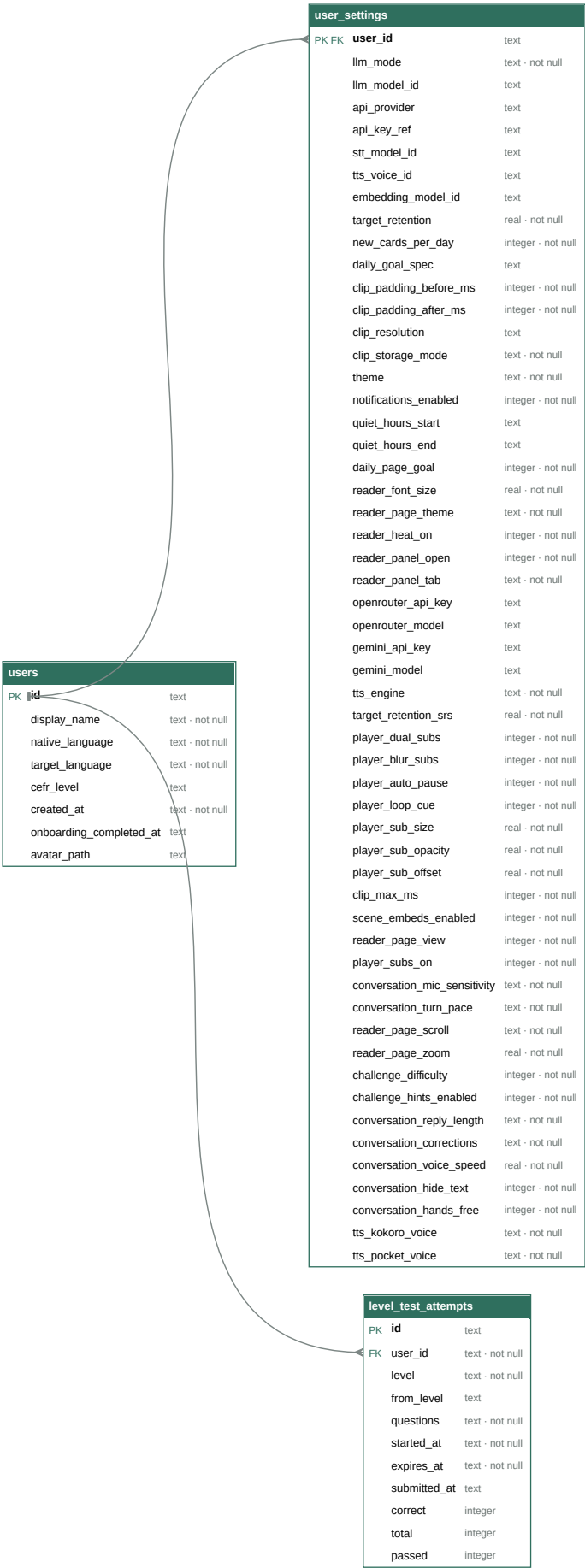
**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table      ➤—| many rows → one row (the line runs from the FK to the key it points at)



# Learners & settings

Who the learner is, every setting they can change, and their level tests. Almost every other table points back to users.

**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table       many rows → one row (the line runs from the FK to the key it points at)



**users** 8 columns

A learner: name, native and target language, CEFR level, when onboarding finished, profile picture.

**user\_settings** 55 columns

One row per learner with every setting: AI engine and keys, voices, reader, video player, review targets, daily goals, notifications.

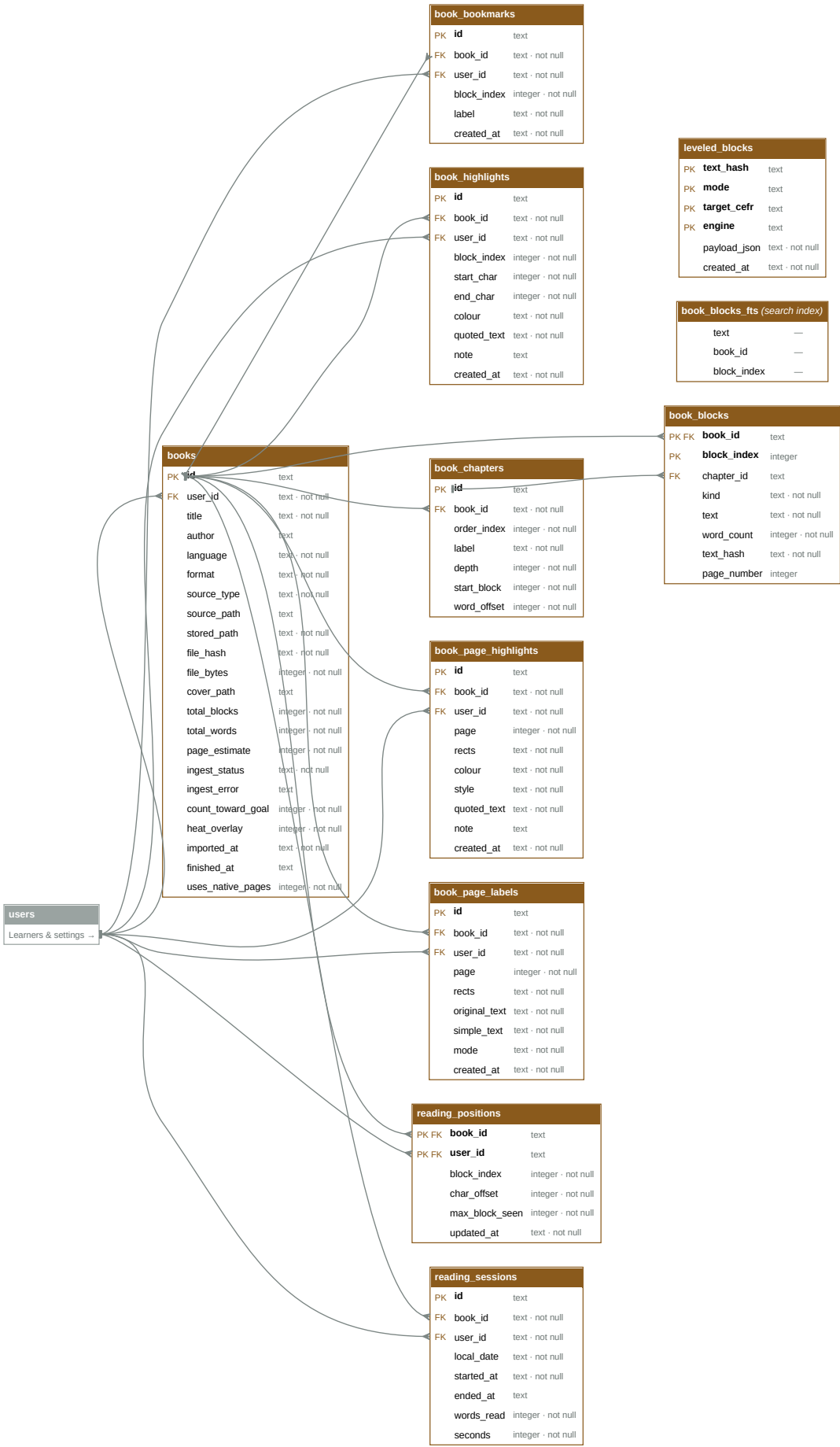
**level\_test\_attempts** 11 columns

Each level test taken: the level, the questions asked, the score, pass or fail, and when it expires.

# Learn by reading

Imported books, split into paragraphs ("blocks"), plus everything the learner does while reading.

**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table      ➞ many rows → one row (the line runs from the FK to the key it points at)



**books** 22 columns

An imported book (TXT, EPUB, MOBI or PDF): the file, title and author, word count and import status.

**book\_chapters** 7 columns

The table of contents: each chapter, its depth and the block it starts at.

**book\_blocks** 8 columns

The book’s text, one row per paragraph, with its word count and page number.

**book\_blocks\_fts** 3 columns

A full-text search index over the book text (SQLite FTS5), used by “search in book”.

**book\_bookmarks** 6 columns

Bookmarks the learner placed.

**book\_highlights** 10 columns

Highlighted passages in flowing text, with colour and an optional note.

**book\_page\_highlights** 10 columns

Highlights drawn on a PDF page, stored as rectangles.

**book\_page\_labels** 9 columns

Simplified-English labels placed over hard text on a PDF page.

**reading\_positions** 6 columns

Where the learner is in each book, and the furthest block they have seen.

**reading\_sessions** 8 columns

Reading per book per day: words read and seconds — feeds the daily goal and streak.

**leveled\_blocks** 6 columns

A shared cache of AI-simplified paragraphs, keyed by the text’s hash and the target level.

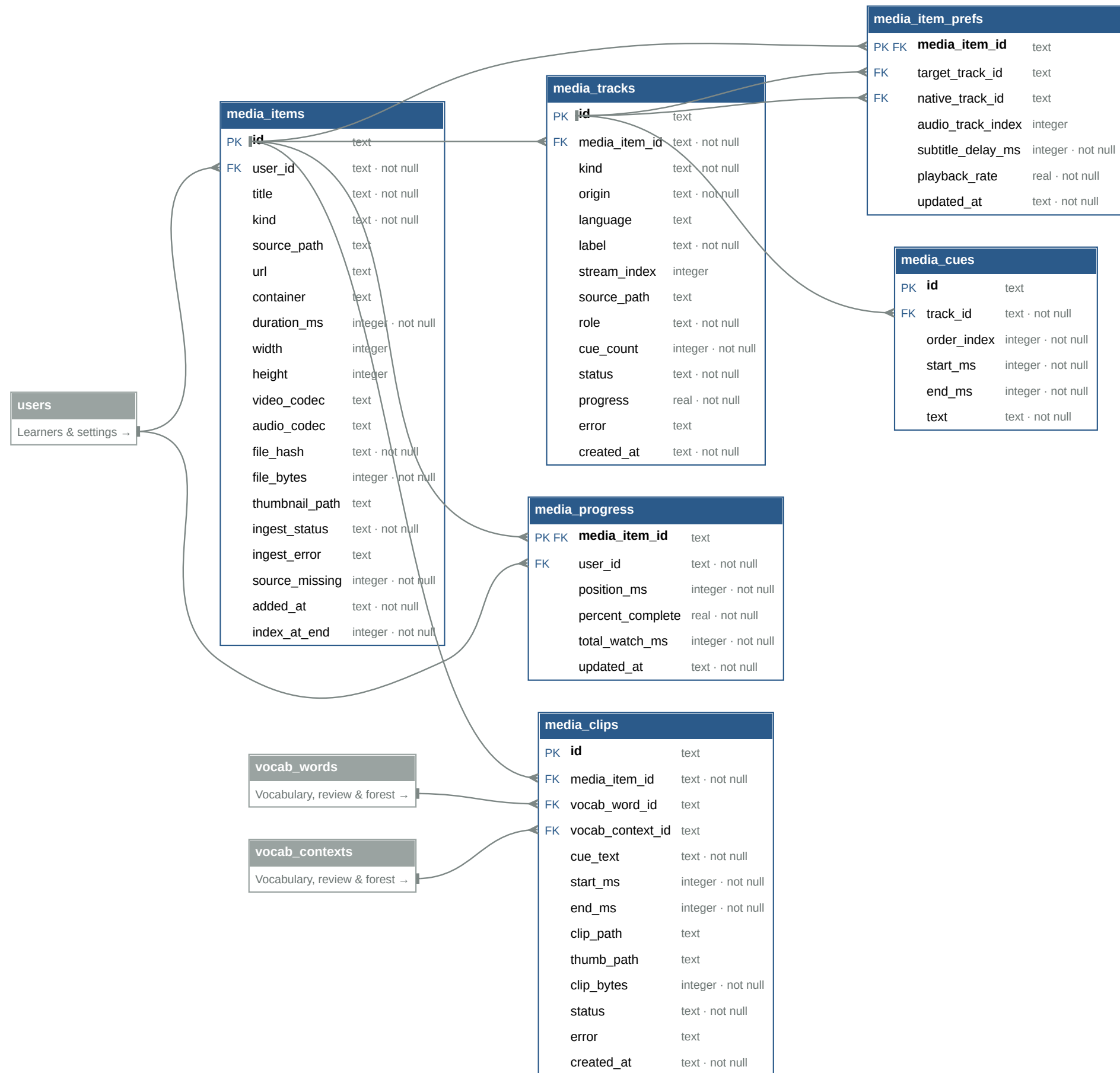
## Learn by watching

Imported videos, their subtitles, and the short clips cut around saved words.

**PK** primary key — identifies each row

**FK** foreign key — points to a row in another table

➤ many rows → one row (the line runs from the FK to the key it points at)



```
media_items 20 columns
```

An imported video or audio file: codecs, size, length and import status.

```
media_tracks 14 columns
```

A subtitle track: embedded in the file, loaded from disk, or generated by speech recognition.

```
media_cues 6 columns
```

One subtitle line: its text and start and end times.

```
media_item_prefs 7 columns
```

Player choices per video: which subtitle tracks, subtitle delay, playback speed.

```
media_progress 6 columns
```

Where the learner stopped, how far through, and total time watched.

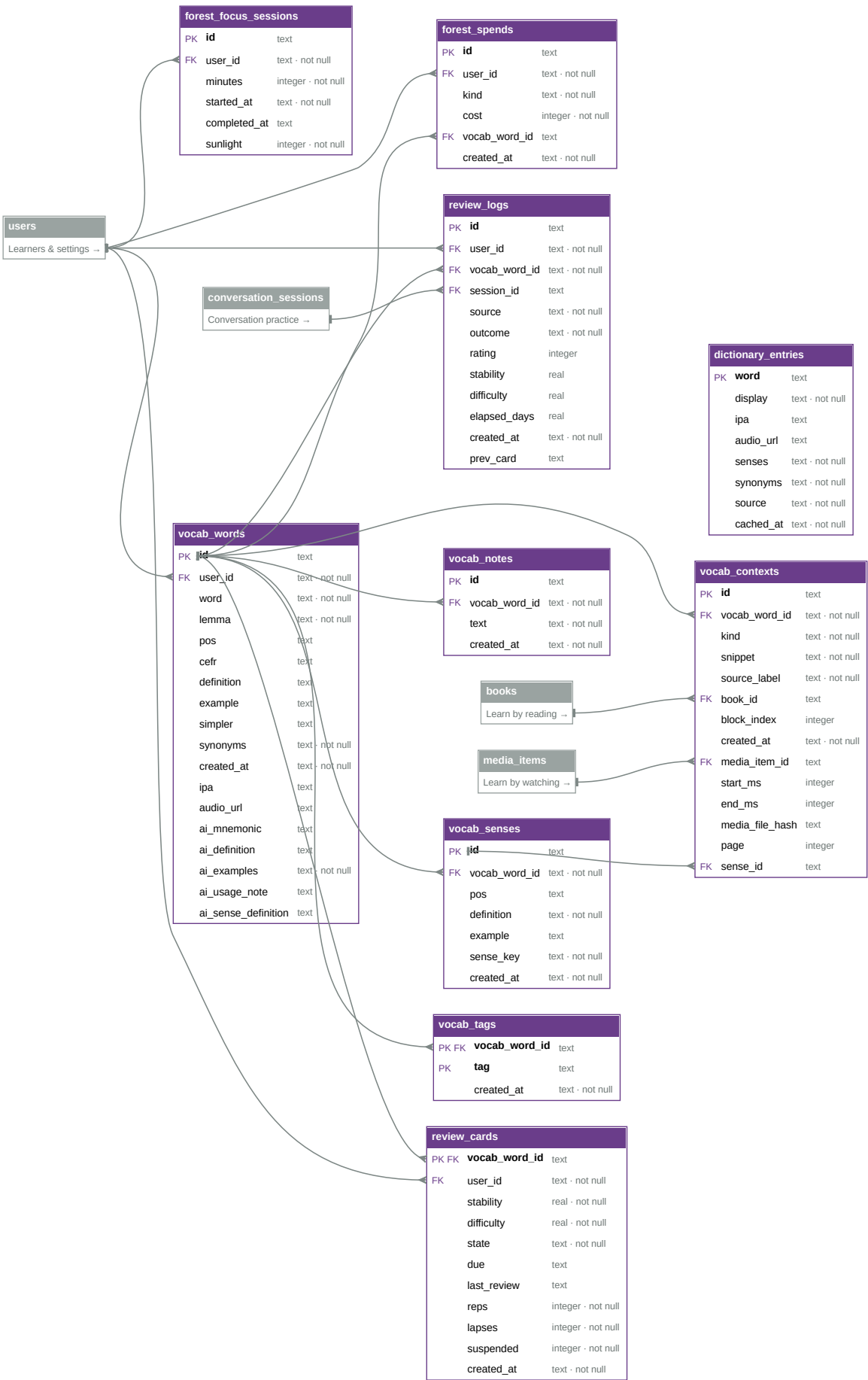
```
media_clips 13 columns
```

A short clip cut around the subtitle line where a word was saved.

# Vocabulary, review & forest

Saved words, where they were met, their flashcards and review history, and the Forest rewards.

**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table      ➞ many rows → one row (the line runs from the FK to the key it points at)



**vocab\_words** 18 columns

A saved word: definition, part of speech, CEFR level, pronunciation and AI-written help.

**vocab\_senses** 7 columns

The meanings (senses) of a saved word.

**vocab\_contexts** 14 columns

Where the word was met: a book page or block, or a moment in a video.

**vocab\_notes** 4 columns

The learner’s own notes on a word.

**vocab\_tags** 3 columns

Tags the learner gives a word.

**review\_cards** 11 columns

The flashcard’s schedule (FSRS): stability, difficulty, state and next due date.

**review\_logs** 12 columns

Every answer ever given — in Review or during a conversation — with the rating and the card’s state before.

**dictionary\_entries** 8 columns

A shared offline cache of dictionary look-ups.

**forest\_focus\_sessions** 6 columns

Focus-timer sessions, which earn “sunlight”.

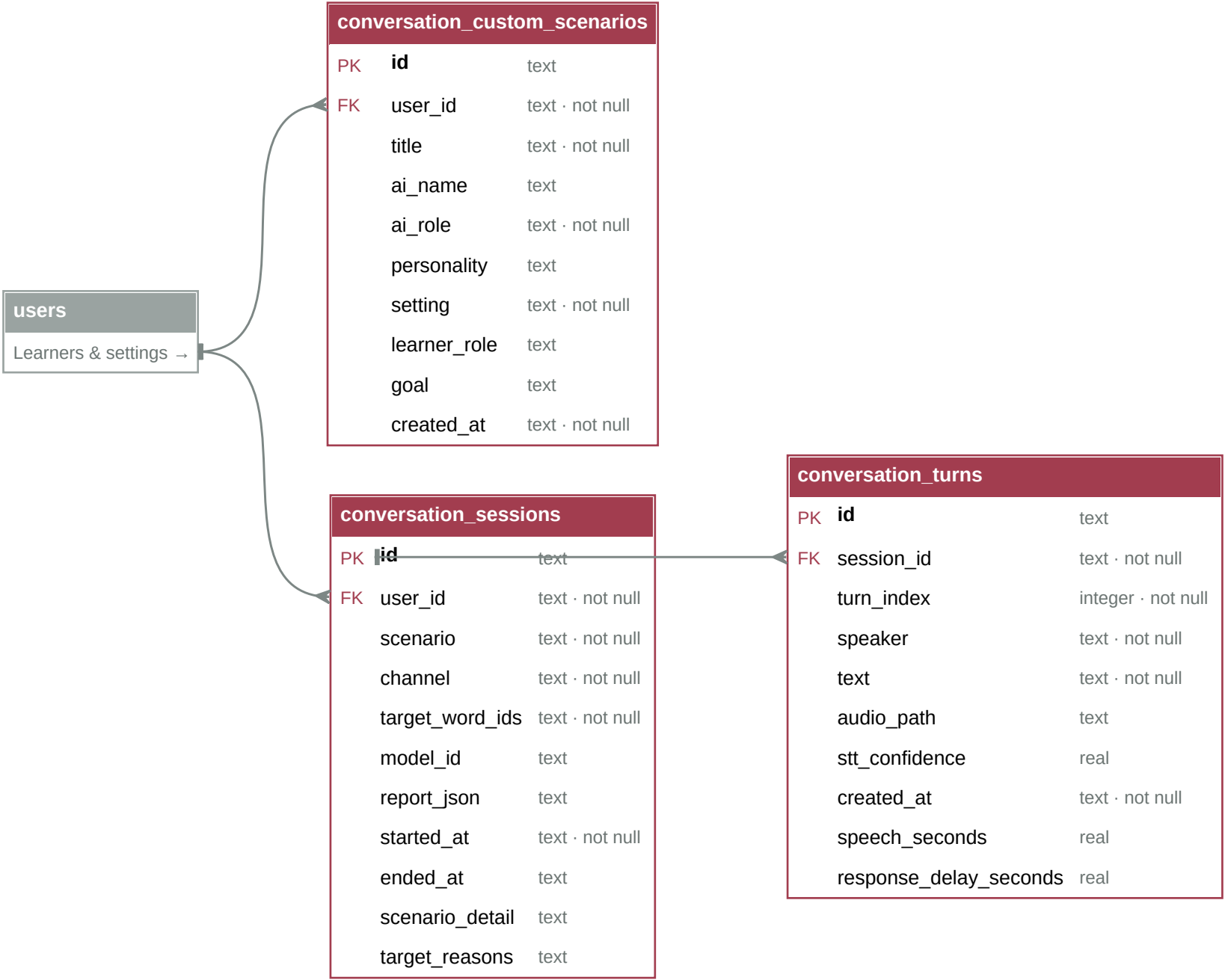
**forest\_spends** 6 columns

Sunlight spent in the Forest, optionally on a particular word.

# Conversation practice

Spoken or typed conversations with the AI partner, turn by turn, and the scenes they happen in.

**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table       many rows → one row (the line runs from the FK to the key it points at)



**conversation\_sessions** 11 columns

One conversation: the scene, the words it asks the learner to use, the AI model, and the report.

**conversation\_turns** 10 columns

One message: who spoke, the text, the recording, speech-recognition confidence and timing.

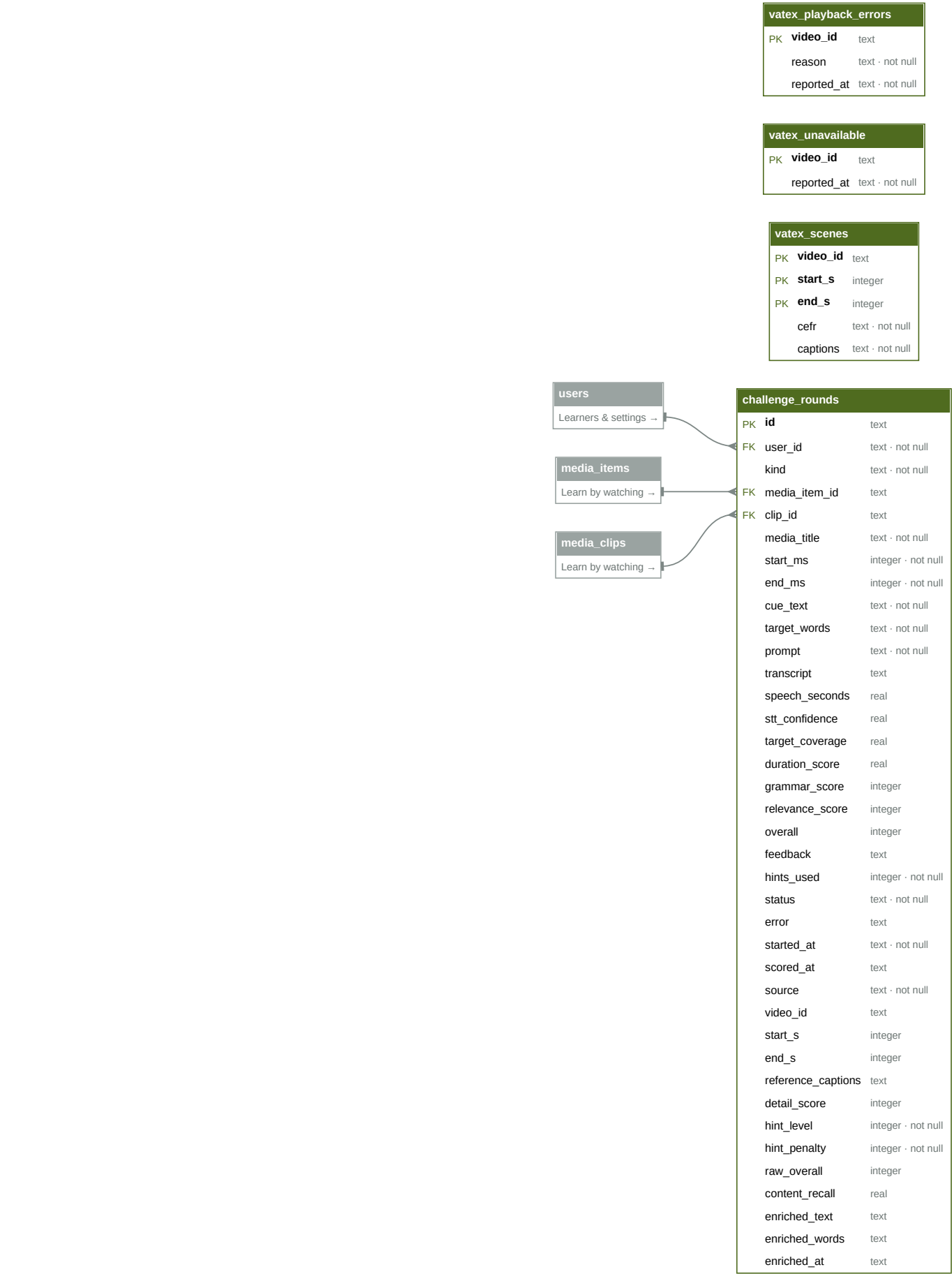
**conversation\_custom\_scenarios** 10 columns

Scenes the learner wrote themselves: the AI's name and role, the setting and goal.

# Scene Description Challenge

Describing a video scene aloud and being scored on it. Scenes come from the learner’s own videos or a built-in library.

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challenge\_rounds 38 columns

One attempt: the clip, what the learner said, each score, hints used and feedback.

vatex\_scenes 5 columns · 34,991 rows built in

The built-in library of about 35,000 short video scenes with reference descriptions (the VATEX dataset).

vatex\_unavailable 2 columns

Library videos found to be unavailable, so they are not offered again.

vatex\_playback\_errors 3 columns

Library videos that failed to play, and why.

# App internals

Bookkeeping for the app itself.

**PK** primary key — identifies each row      **FK** foreign key — points to a row in another table       many rows → one row (the line runs from the FK to the key it points at)

| app_meta |            |      |
|----------|------------|------|
| PK       | <b>key</b> | text |
|          | value      | text |

| _migrations |                 |                 |
|-------------|-----------------|-----------------|
| PK          | <b>filename</b> | text            |
|             | applied_at      | text · not null |

**\_migrations** 2 columns · 44 rows built in  
Which schema migration files have been applied — how the database is upgraded safely between versions.

**app\_meta** 2 columns · 6 rows built in  
App-wide key/value facts, such as one-off data fixes already done.

# How to read this schema

## Tables, rows and columns

A **table** stores one kind of thing — learners, books, saved words. Each **row** is one of them (one learner, one book) and each **column** is one fact about it (a name, a date).

## Keys

A **primary key (PK)** uniquely identifies a row, usually a column called `id`. Some tables use two columns together, e.g. `book_blocks` is identified by `book_id` + `block_index`.

A **foreign key (FK)** stores another row’s primary key, linking the two: `books.user_id` holds the `id` of the learner who imported the book. In the diagrams, each line runs from a foreign key to the key it points at.

## Relationships

Most links are **one-to-many**: one learner has many books; one book has many blocks; one saved word has many contexts. A few are **one-to-one**, where the foreign key is also the primary key: each learner has exactly one `user_settings` row, and each saved word has exactly one `review_cards` row.

## The centre of the design

users is at the centre: almost every table belongs to a learner. The second hub is `vocab_words`: a saved word links to where it was met (`vocab_contexts` → a book or a video), its flashcard (`review_cards`), its review history (`review_logs`) and any video clip cut for it (`media_clips`). That is how reading, watching, conversation and review connect.

## Tables with no links

`dictionary_entries`, `leveled_blocks` and the `vatex_*` tables are **shared caches and libraries**: the same dictionary entry or built-in scene serves every learner, so they belong to nobody.

## How the schema changes

The schema is built by the numbered SQL files in `backend/app/migrations/` (`0001_init.sql` onward). At start-up the app runs any file not yet listed in `_migrations`, so an existing database is upgraded in place without losing data.